SanitiSense AI

Bridging Citizens and Civic Systems Through Intelligent Sanitation Reporting

Team Name: SanitiSense AI

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Team Information

Team Name

SanitiSense AI

Problem Statement

Residents in underserved communities fail to get sanitation hazards (overflowing drains, garbage accumulation) resolved because municipal systems prioritize formal, written, and categorized grievances, ignoring unstructured photo- and voice-based reports.

Team Leader Name

¡Your Name¿

Brief About the Idea

- **SanitiSense AI** is an AI-powered system that enables low-literacy citizens to report sanitation hazards using a photo and an optional voice note
- The system **verifies** the issue using computer vision
- **Extracts contextual urgency** from voice input
- **Converts** informal citizen reports into structured, evidence-backed civic tickets that authorities can act upon

Core Philosophy

Instead of creating another complaint portal, the solution acts as an **intelligent translation layer** between citizen reality and municipal systems.

How Is This Different From Existing Solutions?

Existing Systems:

- Require written, form-based complaints
- Reject or ignore unstructured WhatsApp or verbal reports
- Lack prioritization and aggregation of citizen complaints

Our Differentiation (USP):

- Image-first reporting ensures evidence-based verification
- AI filters spam and invalid submissions automatically
- Converts vernacular, unstructured inputs into structured civic data
- Aggregates repeated issues to expose ignored sanitation hotspots

"We don't help citizens complain — we help systems understand."

How Does the Solution Solve the Problem?

1. Citizens submit a **photo** of a sanitation issue (mandatory)
2. **Optional voice note** provides context such as duration, smell, or obstruction
3. **AI validates** whether the image represents a real sanitation hazard
4. The system **classifies issue type** and estimates severity
5. **Structured and prioritized** sanitation tickets are generated automatically
6. **Repeated complaints** are clustered to identify high-risk zones

Impact

This removes literacy, language, and documentation barriers while enabling accountability.

Features Offered by the Solution

Image Processing:

- Image-based sanitation issue detection and validation
- Automatic spam and irrelevant image rejection
- Severity estimation using visual cues (scale, spread, obstruction)

Voice & Analytics:

- Voice-based context capture in local languages
- NLP-based extraction of urgency indicators
- Clustering of repeated complaints into sanitation hotspots
- Evidence-backed dashboard for authorities or NGOs

Process Flow

Citizen uploads photo of sanitation issue

↓

AI verifies and classifies the image

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Optional voice note adds contextual information

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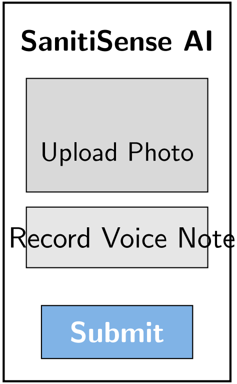
AI structures the issue into a civic ticket

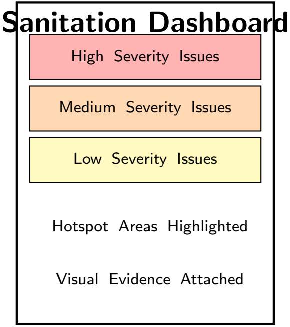
Issues ranked by severity and frequency

Wireframes / Mock Diagrams

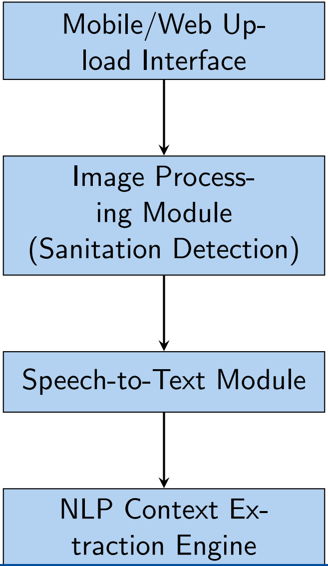
Citizen Interface:



Dashboard Interface:



Architecture Diagram of the Proposed Solution



Technologies to Be Used

- **Computer Vision** for sanitation image classification
- **Speech-to-Text** for local language voice input
- **Natural Language Processing** for context extraction
- **Rule-based + lightweight ML models** for severity scoring
- **Spatial clustering** for hotspot detection
- **Cloud-based backend** for scalability

Estimated Implementation Cost

Cost-Effective Approach

- Uses **lightweight inference models** suitable for low-cost cloud deployment
- **No continuous human moderation** required
- Suitable for **NGO pilots or ward-level rollouts**
- **Pay-per-use or subscription-based** operational cost

Scalable from pilot to city-wide deployment

Expected Impact

Key Outcomes

- **Increased acknowledgment** of valid sanitation complaints
- **Faster response** to high-severity sanitation hazards
- **Reduced repeat complaints** from the same areas
- **Improved accountability** through evidence-backed reporting

Building Trust Between Citizens and Civic Systems

Thank You

Questions?

Team: SanitiSense AI